Applying Navier-Stokes equations to simulate the effects of arterial stenosis on blood flow.

Gabriel K. Staniszewski

Introduction

Even with today's best medicine, stenosis of the arteries still proves to be a challenge for mankind. A simple narrowing of the arteries can have a large effect, like altering the course of blood flow, imparting huge implications on general health.

Arterial stenosis refers to the narrowing of arteries, a network that carries oxygenated blood from the heart to various tissues and organs throughout the body. This vasoconstriction arises from diverse factors, with atherosclerosis being a predominant contributor. Atherosclerosis is the accumulation of plaque along the inner walls of arteries, leading to a reduction in its diameter. This narrowing disrupts the laminar flow of blood, setting the stage for a cascade of physiological consequences.

The repercussions of arterial stenosis exceed more than blocked pipage in our bodies. As blood encounters the obstructed pathways, it faces increased resistance, resulting in altered behavior of blood flow. Reduced blood flow downstream from the stenotic region can compromise the delivery of vital nutrients and oxygen to tissues and organs. This diminished perfusion may lead to ischemia (AHA, 2024), a condition where cells receive insufficient blood supply, potentially culminating in tissue damage or organ dysfunction.

Moreover, the narrowed arteries prompt the heart to exert additional effort in pumping blood through the constricted pathways. This increase of workload may contribute to a rise in blood pressure, exerting strain on both the heart and the entire cardiovascular system. Arterial stenosis thus can be thought as a turning point in the workings of blood circulation and cardiac function.

The dangers of arterial stenosis depend upon the location and severity of the narrowing. In coronary arteries, stenosis can lead to angina (AHA, 2024), chest pain resulting from insufficient blood supply to the heart muscle. Severe stenosis may cause myocardial infarction, commonly known as a heart attack, as the blood flow to a portion of the heart muscle is abruptly cut off.

Similarly, stenosis in cerebral arteries may manifest as transient ischemic attacks (TIAs) or strokes (AHA, 2024), depending on the degree of blood flow compromise. Peripheral arterial stenosis can result in claudication, characterized by pain or cramping in the limbs during physical activity, reflecting insufficient blood supply to the extremities.

Several factors contribute to the development and progression of arterial stenosis. Atherosclerosis, often driven by elevated cholesterol levels, hypertension, and smoking, serves as a primary precursor. Genetic predispositions, diabetes, and inflammatory conditions also contribute to the initiation and acceleration of arterial narrowing. Lifestyle choices, including diet, physical activity, and stress management, further influence the risk of developing stenotic lesions (Penn, 2024).

In my investigation, this phenonemon will be evaluated by data obtained from solving the Navier-Stokes equations for parameters found in the carotid artery. In practical applications, such as simulating blood flow in arteries, solving the full Navier-Stokes equations for every particle in the fluid becomes impractical due to obvious complexity. Instead, simplified models and average values are often employed to approximate the behavior of the fluid system. This approach balances computational efficiency with accuracy, enabling researchers to study phenomena like pulsatile flow – characteristic of blood flow driven by the heart's rhythmic contractions – with reasonable utilization. In this instance, Simscale was used as my computational fluid dynamics (CFD) software of choice.

Aim

The aim of this exploration is to apply the Navier-Stokes equations in a fluid dynamics program to simulate stenosis of the carotid artery, that being the artery that supplies our brain, eyes and jaw, and use the data to create a model of the effect of stenosis percentage on blood flow velocity. Next, analyze why this relationship behaves in such a way and calculate the derivative to further understand the underlying mechanisms from a mathematical point of view. In order to make this aim feasible we have to make two assumptions in the Navier-Stokes equations; first that the fluid is incompressible, meaning its volume will remain constant and second, that the system is isothermal (no heat is lost by the system). Usually a third assumption is made that the simulated fluid (in this case blood) is Newtonian, which means that the shear rate is directly proportional to the shear stress (Lever, 2005). However, through the use of the Bird-Carreau model (which will be discussed later), we can simulate blood as a non-Newtonian fluid, increasing the accuracy of the model.

Rationale

The human heart, with its intricate design and profound role as the guardian of life, has captivated my fascination from an early age. My aspiration to pursue a career in medicine, particularly as a cardiothoracic surgeon, is rooted in the profound admiration I hold for this organ. It is, in my opinion, the most beautiful creation of the world – the vessel from which human life emanates. This deep appreciation for the heart, coupled with my affinity for mathematics, has inspired me to embark on an exploration into the intersection of medicine and mathematical modeling, with a specific focus on arterial stenosis.

As I envision my future as a cardiothoracic surgeon, I recognize the importance of understanding not only the anatomical intricacies of the heart but also the dynamic physiological processes that govern its function. Arterial stenosis, a prevalent and impactful cardiovascular condition, presents a unique opportunity to merge my two passions – medicine and mathematics.

Exploring arterial stenosis allows me to dive into the complexities of blood flow dynamics, a crucial aspect of cardiovascular health. Through mathematical modeling and simulation, I aim to graph the blood flow in the carotid artery under different stenosis conditions. Through this exploration I wish to bring a deeper understanding into the specific effect of narrowing of the arteries on the hemodynamic (dynamics of the blood) patterns within the arterial system, providing valuable insights into the physiological consequences of stenosis.

The application of derivatives in this exploration is intentional. Derivatives become the method through which I can analyze and quantify the variations in blood flow rates along the stenotic region. By experimenting with the dynamic changes caused by stenosis using mathematics, I can broaden my understanding of the challenges posed by arterial stenosis and its impact on the cardiovascular system.

In summary, the convergence of paths of my admiration for the heart, my aspiration to pursue medicine, and my affinity for mathematics has led me on this exploration into arterial stenosis. By graphing blood flow in the carotid artery and understading the hemodynamic changes with derivatives, I aim to contribute to the growing body of knowledge that bridges the domains of medicine and mathematics, ultimately preparing me for the journey towards becoming a cardiothoracic surgeon.

Describing the situation

The Navier-Stokes equation is a cornerstone of fluid dynamics, encapsulating the behavior of fluid flow in a given domain. It combines Newton's second law with the conservation of momentum and the viscous effects within the fluid. While most variables in the Navier-Stokes equations remain constant, more can be added or removed based on the physics being simulated (Simscale, 2023). For example, if heat transfer, phase change, or chemical reactions need to be considered, more terms will be introduced into the governing equations.These equations are notoriously challenging to solve analytically, particularly for complex, three-dimensional flows. As a result, numerical methods are often employed to approximate solutions, providing insights into phenomena ranging from atmospheric dynamics to blood flow in arteries.

The continuity equation (Bistafa, Sylvio, 2017):

$$\nabla \cdot V = 0$$

where:

∇ is the divergence,

V is a vector field (representing the velocity of particles).

The continuity equation, stemming from the principle of mass conservation, complements the Navier-Stokes equation by ensuring that the flow field maintains a consistent mass balance. This equation asserts that the divergence of the velocity field is zero, implying that mass is neither created nor destroyed within the fluid domain. Essentially, it states that what flows into a given region must flow out, preserving the overall mass within the system. This principle underpins various fluid dynamics simulations, ensuring the accuracy and fidelity of the results obtained.

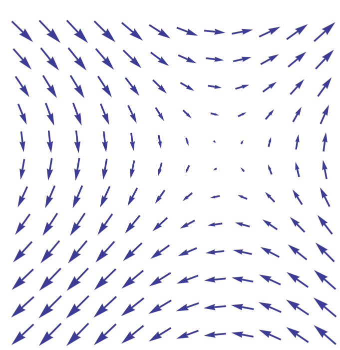


Figure 1: A simple, 2D vector field [17]

Vector fields play a crucial role in describing fluid behavior, representing quantities such as velocity, pressure, and force as vectors distributed throughout the fluid domain. These fields elucidate how these quantities vary spatially and temporally, providing insights into the complex dynamics of fluid flow. Divergence, a fundamental concept in vector calculus, quantifies the extent to which a vector field diverges or converges at a given point. In fluid dynamics, divergence serves as a measure of how fluid is either accumulating or dispersing within a particular region, influencing the overall flow patterns.

Newton's second law:

$$\sum F = m \cdot a$$

Mass can be described as density, while acceleration can be written as the derivative of the change of velocity over time, which incorporates the vector field when written as a partial derivative (Bistafa, Sylvio, 2017):

$$m = \rho$$

$$a = \frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\partial V}{\partial t} + V \cdot \nabla V$$

Thus, the sum of forces we usually consider (pressure, friction, gravity) gives us the equation:

$$\sum \text{Forces} = \text{pressure gradient } (-\nabla \rho) + \text{friction } \left(\mu \nabla^2 V\right) + \text{external, usually gravity } (\text{F}, \rho \cdot g)$$

So Newton's 2nd law can be rewritten as:

$$\rho \frac{\mathrm{d}V}{\mathrm{d}t} = -\nabla \rho + \mu \nabla^2 V + \rho g$$

Otherwise known as the momentum equation (Bistafa, Sylvio, 2017).

The momentum equation, derived from Newton's second law, expresses the balance between inertial forces, pressure gradients, and viscous effects within the fluid. It explains how the velocity field evolves over time in response to external forces and internal interactions, dictating the fluid’s motion and behavior. By rewriting Newton’s second law in terms of fluid dynamics, the momentum equation provides a detailed description of the forces acting on the fluid and their resultant effects on its motion, in this case: of blood.

The representation of blood flow in arteries involves determining key parameters such as blood viscosity, density, and flow velocity, which influence the fluid dynamics within the cardiovascular system. Experimental measurements and numerical studies serve as crucial validation tools, ensuring that computational models accurately reflect real-world phenomena. For instance, studies employing laminar flow models and advanced turbulence models like Menter’s hybrid k-ε/k-ω Shear Stress Transport model validate their successfulness in understanding the complexities of pulsatile blood flow.

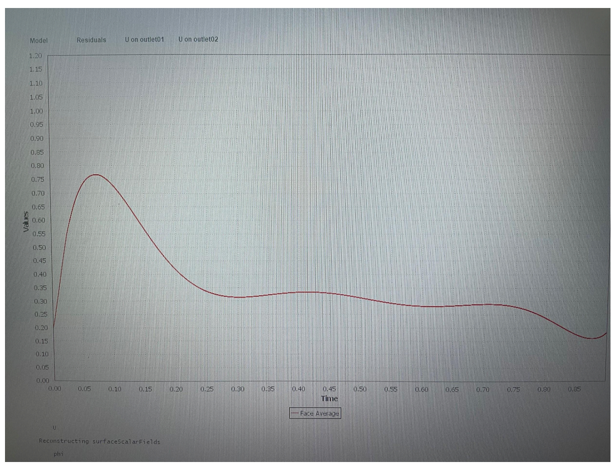


Figure 2: Pulsatile flow data of the blood caused by the pumping of the heart used during my simulations from (Tash et. al, 2012).

Pulsatile flow in arterial blood flow, driven by the cyclic pumping action of the heart, presents unique challenges and implications for cardiovascular simulations. The pulsatile nature of blood flow introduces fluctuations in velocity, pressure, and shear stress within the arteries, affecting factors such as wall shear stress and turbulence intensity. Understanding and accurately simulating pulsatile flow dynamics are essential for predicting phenomena like arterial plaque formation and assessing the success rate of medical interventions.

In summary, the mathematical foundation of fluid dynamics, which encompasses concepts such as the Navier-Stokes equation - the continuity equation and momentum equation, provides a decent playground for studying complex fluid phenomena. Through numerical simulations and validation studies, researchers can gain insights into diverse applications, from cardiovascular dynamics to atmospheric circulation of hurricanes, advancing our understanding of fluid behavior and its implications for various fields.

Method

The initial methodology for this exploration involved the use of SimFlow, a computational fluid dynamics software, to simulate and analyze the effects of arterial stenosis on blood flow dynamics within the carotid artery.

Acquiring accurate medical data is the initial step in constructing a realistic virtual model. Dimensions, curvature, and initial blood flow conditions are used from relevant medical sources, ensuring the accuracy of the representation. The precision of this data is crucial for a faithful reconstruction of the carotid artery within the simulation.

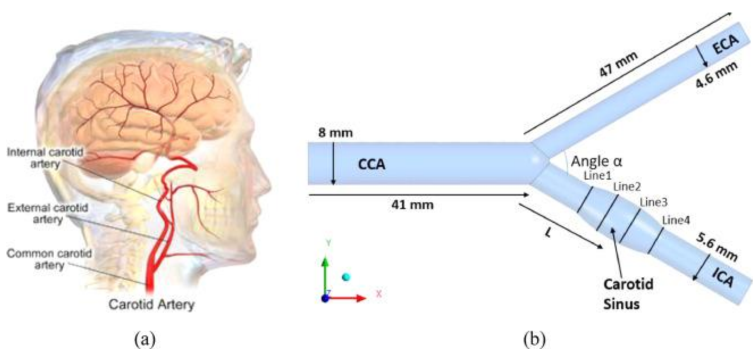


Figure 3: Data used to construct the virtual model of the carotid artery (Muthusami, 2013)

The next stage involves the implementation of acquired medical data into SimFlow. This process transforms the data into a 3D model, capturing the intricate details of the carotid artery. The goal is to create a virtual environment that replicates the actual anatomical features like the features of the inner arterial wall, or the characteristic of the blood itself like its physical properties.

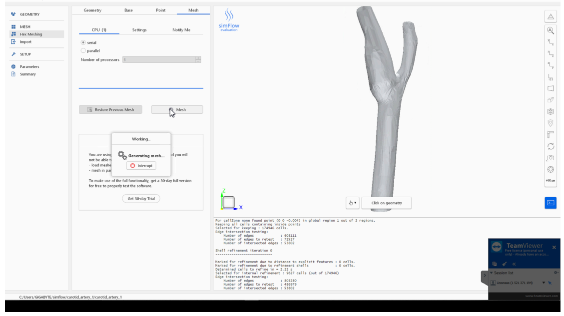
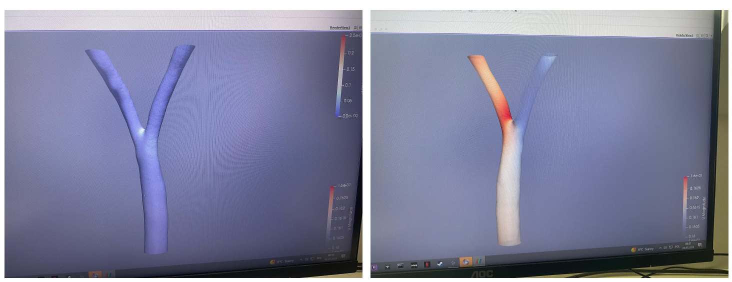


Figure 4: 3D mesh of the carotid artery created in SimFlow

To simulate the impact of arterial stenosis, the model is designed to incorporate variable stenosis levels. This feature allows for the introduction of different degrees of narrowing within the artery. Geometries are adjusted to simulate stenosis at 0% to 100% at 10% intervals, covering a spectrum of clinical scenarios and enhancing the insights of the simulation.

Next comes the simulation itself. Through the simulation we collect vital data for the analysis of this investigation, involving the systematic recording of key parameters at different points along the stenotic region. Flow velocity, pressure gradients, and turbulence patterns are documented under each stenotic condition.



Figures 5 & 6: Two separate results from differently averaged input data.

While repeating my measurements in SimFlow under slightly different conditions I could not replicate my results as seen in figures 5 and 6. Which was a huge problem. The simulation wasn't reliable under idealized conditions so I had to conclude that the program I was using wasn't reliable for this sort of application. This resulted in me deciding on using Simscale instead, after conducting more research. In addition, it was at that point that I switched to a

$k - \Omega$ SST turbulence model after finding research that praised its reliability for cardiovascular application (Tan, 2008).

Analyzing the data involves the application of mathematical derivatives to understand the rate of change of parameters of flow velocity following different stenotic levels, revealing patterns and trends in the hemodynamics.

Part 1: Solving the Navier-Stokes equations

(by implementing computational fluid dynamics (CFD))

To begin the process of solving the Navier-Stokes equations, several steps are involved, starting with defining the system geometry, boundary conditions, and fluid properties. In the context of simulating blood flow through stenotic arteries, accurate representation of the arterial geometry, including dimensions and curvature, is crucial. Additionally, boundary conditions, such as prescribed inlet pressure and no-slip conditions on the arterial walls, need to be specified. These boundary conditions are represented mathematically as constraints applied to the velocity and pressure fields:

- The no-slip boundary condition is expressed as $\boldsymbol{V} = 0$ at solid boundaries, where $\boldsymbol{V}$ represents the fluid velocity vector.
- Inlet conditions, such as the prescribed inlet pressure of $P_{inlet} = 1184 \text{ Pa}$, are defined to initialize the flow field within the computational domain.

Furthermore, fluid properties, including viscosity and density, play a crucial role in determining the behavior of the fluid. In the simulation of blood flow, a non-Newtonian viscosity model, such as the Bird-Carreau model, may be employed to account for the shear-thinning behavior of blood (Simscale, 2022). This is all done by the software, after inputting data from (Lynch, 2018). The model is represented by the following equation:

The Bird-Carreau model (Simscale, 2022):

$$\mu = \frac{\mu_\infty + (\mu_0 - \mu_\infty)(1 + (\lambda\dot{\gamma})^a)^{(n-1)/a}}{1 + (\lambda\dot{\gamma})^a}$$

where:

- μ is the dynamic viscosity.
- μ_0 and μ_∞ are the viscosity at zero and infinite shear rates, respectively.
- λ, n, a and are empirical parameters.
- $\dot{\gamma}$ represents the shear rate.

Additionally, the density of the fluid, denoted by ρ, is specified as 1060 kg/m^3 (Lynch, 2018).

Incorporating turbulence modeling is another essential aspect of solving the Navier-Stokes equations, especially in complex flow scenarios. The $k-\Omega$ SST turbulence model is commonly used for its ability to accurately predict turbulence in a wide range of flow conditions. This model combines the benefits of the $k-\varepsilon$ and $k-\omega$ models, providing robust predictions of turbulent flow phenomena (CFDonline, 2011).

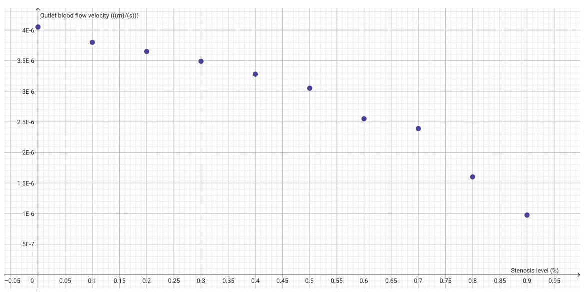
Putting all of these starting conditions and equations into the CFD software results with this dataset:

Stenosis level (%)	0	10	20	30	40
Outlet flow rate	$4.05 \cdot 10^{-6}$	$3.80 \cdot 10^{-6}$	$3.65 \cdot 10^{-6}$	$3.49 \cdot 10^{-6}$	$3.28 \cdot 10^{-6}$
Stenosis level (%)	50	60	70	80	90
Outlet flow rate	$3.05 \cdot 10^{-6}$	$2.55 \cdot 10^{-6}$	$2.39 \cdot 10^{-6}$	$1.6 \cdot 10^{-6}$	$0.976 \cdot 10^{-6}$

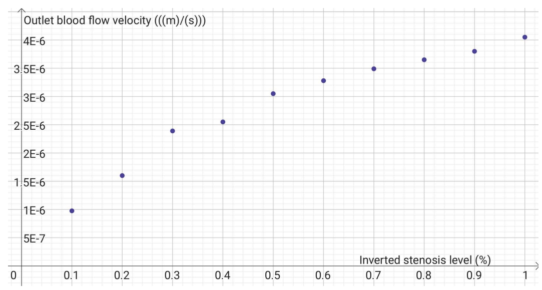
Table 1: Containing the data of the results of different simulations of stenosis from 0-90%.

Part 2: Differentiating the velocity relationship

In the analysis of fluid dynamics, specifically in modeling blood flow through stenotic arteries, it is essential to differentiate velocity to understand the hemodynamic implications of arterial narrowing. This process involves creating a model function that captures the interplay between velocity and stenosis. Here, I delve into the method of creating such a model function and evaluating its implications based on external research findings.



Graph 1: Plotting the outlet blood velocity against stenosis level.



Graph 2: Plotting the outlet blood velocity against inverted stenosis level.

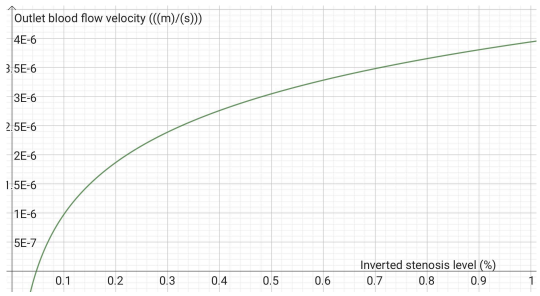
When the data is represented on graph 1, I noticed that it could be modeled by a logarithmic function mirrored on the y-axis. In order to do that, I first had to invert the percentage values of the stenosis level in the dataset from (x) to **X**, where $X = 1 - x$ as shown on graph 2. This inversion allows us to create a natural logarithmic growth model that describes how blood flow velocity changes with varying degrees of stenosis in a graphic display calculator (GDC). The natural logarithmic growth model calculated by the GDC is expressed as:

$$y = 3.941 \cdot 10^{-6} + 1.288 \cdot 10^{-6} \cdot \ln(X)$$

where:

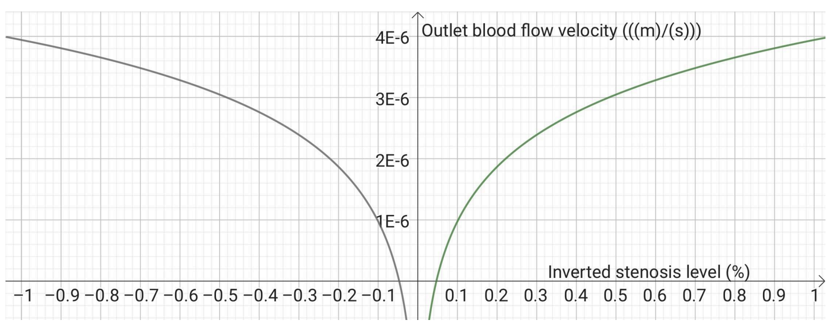
- y represents the outlet blood flow velocity,
- X is the inverted stenosis percentage.

The domain of this function is $0 \leq X \leq 1$, representing the range of inverted stenosis percentages, and the range is $0 \leq y \leq 4.3 \cdot 10^{-6}$, representing the range of blood flow velocities.



Graph 3: Logarithmic model of blood flow velocity and stenosis derived from simulated results.

In order to be able to analyze the effects of stenosis on outlet blood velocity, we need to revert the domain to true stenosis percentage. To achieve this we first must mirror the function about the y-axis (y = -***X***). This reflection draws a natural logarithmic model, which decreases as the x-axis grows, reflecting how blood flow velocity decreases as stenosis percentage increases.

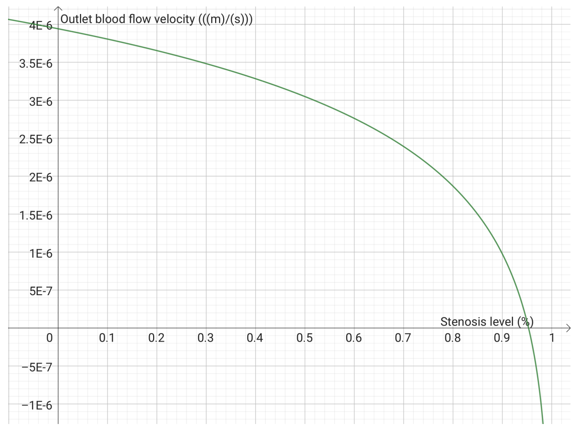


Graph 4: Mirrored function of the logarithmic model (green) to a decreasing model (gray).

The reflection moves the function outside of the domain, so we need to add 100% to $-X$. The flipped function is given by:

$$y = 3.941 \cdot 10^{-6} + 1.288 \cdot 10^{-6} \cdot \ln(-X + 1)$$

By flipping the function, we can evaluate the behavior of blood flow velocity as stenosis percentage increases, providing insights into the hemodynamic consequences of arterial narrowing.



Graph 5: Vertically mirrored logarithmic model, translated by 1 unit to the right, to insert the function into the domain.

The simulated conditions indicate critical thresholds of stenosis severity and their physiological implications. For example, at 95.3% stenosis, blood flow through the stenosed pathway ceases altogether (assuming that the heart doesn't begin to compensate with higher pressure), highlighting the critical nature of severe arterial narrowing. Furthermore, at 78.3% stenosis, blood velocity decreases by half, indicating a significant reduction in flow rates. These findings serve as benchmarks for evaluating the impact of stenosis on blood flow dynamics.

Based on the simulated findings, specific stenosis percentages associated with significant decreases in blood velocity can be determined. For instance:

- A 25% decrease in blood velocity corresponds to a stenosis percentage of approximately 53.7%.
- A 50% decrease in blood velocity corresponds to a stenosis percentage of approximately 78.3%.
- A 75% decrease in blood velocity corresponds to a stenosis percentage of approximately 90.2%.

The 50% decrease in blood velocity corresponding to a stenosis percentage of approximately 78.3%, is a critical threshold indicating severe reduction in perfusion (Guarini, 2012).

By incorporating these physiological thresholds into our analysis, we gain a deeper understanding of the hemodynamic consequences of arterial stenosis. Specifically, we can relate the degree of stenosis to critical physiological effects such as blood flow cessation and hypotension. This information is crucial for clinical decision-making, as it helps identify patients at risk of severe complications and guides appropriate interventions to mitigate the impact of arterial narrowing. Additionally, it highlights the importance of early detection and management of vascular diseases to prevent adverse outcomes associated with hemodynamic instability.

Through the evaluation of stenosis percentages, researchers gain insights into the relationship between arterial narrowing and hemodynamic alterations, providing valuable information for clinical assessment and treatment planning. The analysis enhances our understanding of the complex relationship between stenosis severity and blood flow dynamics, aiding in the development of effective diagnostic and therapeutic strategies for patients with vascular diseases.

Differentiation

The analysis of simulated blood flow data within the stenotic region involves employing mathematical derivatives to observe the rate of change in parameters, specifically focusing on flow velocity.

By utilizing derivatives, we can precisely examine how flow velocity evolve along varied stenosis of the carotid artery. Derivatives provide a quantitative measure of the rate of change, enabling a detailed exploration of how these crucial parameters respond to varying degrees of arterial narrowing.

Let V represent flow velocity. The derivative, $\frac{dV}{dx}$, indicates the rate of change of velocity in respect to stenosis severity (x).

$$[\frac{dV}{dx} = \lim_{\Delta x \to 0} \frac{\Delta V}{\Delta x}]$$

These derivatives provide insights into how velocity change over increasing stenosis blockage of the artery, offering a quantitative understanding.

$$V(x) = 3.9408932520002 \cdot 10^{-6} + 1.287646959763 \cdot 10^{-6} \cdot \ln(-X + 1)$$

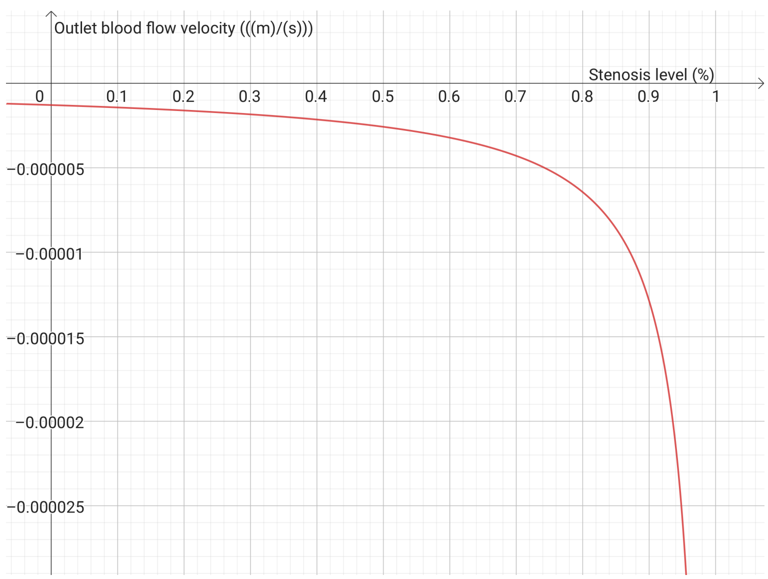
$$\frac{dV}{dx} = \frac{d}{dX}\left(3.940893250002 \cdot 10^{-6} + 1.287646959763 \cdot 10^{-6} \cdot \ln(-X + 1)\right) =$$

$$= \left(1.288 \cdot 10^{-6} \cdot \frac{1}{(-X + 1)}\right)(-1) =$$

$$= -\frac{1.288}{10^6} \cdot \frac{1}{(-X + 1)} =$$

$$= -\frac{1.288}{10^6(-X + 1)} =$$

$$= -\frac{1.288 \cdot 10^{-6}}{-X + 1}$$



Graph 6: Graphical representation of the derivative of flow velocity and stenosis ($\frac{dV}{dx}$)

Analysis

The implications of the results obtained from the analysis of arterial stenosis and its effects on blood flow dynamics are varied and extend across different domains, including clinical medicine, biomedical engineering, and patient care. I will take some time to expand the implications of the results in these fields.

Starting with the mathematical implications; the derivative of velocity over stenosis level is negative over the entire domain. However, as can be seen on graph 6, at around 50% stenosis, the negative derivative of velocity has doubled. This stenosis thus is already quite dangerous as one would most likely not notice any symptoms at first, but with the progression of the disease - severe symptoms will probably quickly arise. This is additionally supported by the graph when we look at the value of the derivative of around 75%, which is about 4 times greater than the starting value of $x = 0$. As previously mentioned, around this stenosis level a critical threshold of persuasion is reached with severe health risks (Guarini (2012).

Moving on to clinical diagnosis and management; the results shed light on the critical thresholds of arterial stenosis severity that warrant clinical attention. Identifying specific stenosis percentages associated with significant decreases in blood flow velocity allows healthcare professionals to better diagnose and monitor patients with vascular diseases. Doctors can use this information to stratify patients based on the severity of arterial narrowing and prescribe treatment strategies accordingly. For instance, patients approaching the critical threshold where blood flow ceases altogether may require urgent intervention, such as surgical revascularization or angioplasty, to restore blood flow and prevent ischemic complications.

The findings also have direct implications for treatment planning and intervention strategies in patients with arterial stenosis. By quantifying the degree of blood flow reduction at different stenosis percentages, clinicians can make informed decisions about the most appropriate course of action for individual patients. For example, patients with moderate stenosis may benefit from lifestyle modifications and pharmacological interventions aimed at reducing cardiovascular risk factors and improving blood flow instead of risking life-threatening surgery. In contrast, patients with severe stenosis approaching critical thresholds may require more aggressive interventions, such as surgical revascularization or endovascular procedures, to restore blood flow and prevent adverse outcomes.

The ability to quantify the hemodynamic consequences of arterial stenosis allows for better risk assessment and prognostication of patients with vascular diseases. By correlating specific stenosis percentages with clinical outcomes, such as myocardial infarction, stroke, or mortality, clinicians can identify high-risk patients who may benefit from closer monitoring and proactive management. Additionally, the findings provide valuable prognostic information for guiding discussions with patients about their disease prognosis and treatment options, empowering them to make informed decisions about their healthcare.

In conclusion, the implications of the results obtained from the analysis of arterial stenosis and its effects on blood flow dynamics covers many fields and hold significant promise for improving clinical outcomes as well as enhancing patient care. By leveraging these insights, clinicians, researchers, and engineers can collaborate to develop more effective strategies for diagnosing, treating, and managing arterial stenosis, ultimately improving the lives of patients with vascular diseases.

Evaluating results

In the context of arterial stenosis, a 50% reduction in blood flow represents a substantial impairment in vascular function that can have profound implications for patient health. Clinically, this degree of stenosis often serves as a critical threshold for intervention (Louridas, 2005), with treatment decisions guided by the severity of blood flow restriction.

By focusing on the stenosis percentage corresponding to a 50% reduction in blood flow, we can assess the model's ability to accurately predict clinically significant outcomes. Real data: Stenosis percentage associated with a 50% reduction in blood flow = 85% (Guarini, 2012) My model: Predicted stenosis percentage associated with a 50% reduction in blood flow = 78%

Using the following equation for percentage error;

$$Percentage\ error = \left| \frac{Real\ Data\ -\ Model\ Prediction}{Real\ Data} \right| \cdot\ 100\%$$

I can calculate the percentage error of my model:

$$Percentage\ error = \left|\frac{85-78}{85}\right| \cdot 100 = \left|\frac{7}{85}\right| \cdot 100\% \approx 8.24\%$$

The calculated percentage error of approximately 8.24% between the predicted stenosis percentage and the value reported in real data is a matter of concern but also offers valuable insights into the model’s performance. I recognize that while it successfully identified a stenosis percentage associated with a significant reduction in blood flow, there are areas for improvement to increase its accuracy and usability.

As to the strengths of my model, while developing the model, I adopted a quantitative approach, using mathematical modeling to investigate the dynamics of arterial stenosis and blood flow reduction. This methodology led to a structured analysis of possible physiological responses to arterial narrowing, allowing for the identification of critical thresholds indicative of hemodynamic compromise. Moreover, the model demonstrated proficiency in pinpointing these critical thresholds, offering valuable insights into disease progression and aiding in clinical decision-making. By quantifying the extent of blood flow reduction across varying degrees of stenosis, the model provides information for healthcare providers, enabling them to group patients based on disease severity and prescribe treatments accordingly.

Despite its strengths, the model exhibited inconsistencies in accuracy, evidenced by the observed percentage error. This gap between predicted stenosis percentages and real-world data suggests the presence of unaccounted factors or inaccuracies within the model’s formulation, thereby compromising its reliability and applicability in clinical settings. Furthermore, the model’s sensitivity to underlying assumptions emerged as a significant limitation, as variations in arterial geometry, blood viscosity, and patient-specific hemodynamic parameters could exert undue influence on model outcomes. These factors highlight the need for a more nuanced approach to modeling arterial stenosis and blood flow dynamics.

Moving forward, I recognize several opportunities for enhancing the model's performance and addressing its inherent weaknesses. First and foremost, strict validation against empirical data from clinical studies and experimental observations is crucial to improve the model's predictive accuracy and establish its credibility within the scientific community. Additionally, the integration of additional physiological variables, such as changes in vessel compliance (Klabunde, 2024), endothelial function (Félétou, 2011), or collateral circulation (Faber, 2014), holds promise for enriching the model's predictive capacity. By accounting for these extraneous variables, the model can better capture the complexity of hemodynamic responses to arterial stenosis and provide more accurate prognostic information. Furthermore, refinement of the model's methodology, including sensitivity analyses and adjustments to underlying assumptions, is essential to limit sources of error and improve its overall accuracy. By investigating model components and continuously validating against real-world data, the model's utility as a tool for shedding light on the complexities of arterial stenosis and informing clinical decision-making can be increased.

In conclusion, while my model demonstrates noticeable strengths in its quantitative approach and critical threshold identification, there exist notable opportunities for improvement. By addressing weaknesses related to predictive accuracy and assumption sensitivity through external validation, integration of more extraneous variables, and refinement of my methodology, a more reliable and clinically relevant model for assessing arterial stenosis and guiding patient care could be developed.

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